

WILL BORDEN

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EDUCATION

The University of Melbourne

Feb 2022 – Expected Dec 2025

B.S. in Electrical Engineering Systems – Weighted Average Mark: 87.3 / 100 (First Class Honors)

Diploma in Languages – French

The University of Bordeaux

Aug 2024 – Dec 2024

Completed coursework towards French language diploma

Academic exchange through the University of Melbourne

EXPERIENCE

Melbourne Space Laboratory

Jul 2025 – Present

Research Assistant - Electronics Engineering

Melbourne

- Designed and validated thermal control unit for an Australian Space Agency lunar rover, carrying a NASA payload to the southern lunar pole.
- Implemented an NVIDIA Jetson-based space-grade camera system, launching on a Gilmour Space satellite in June 2026.
- Designed and manufactured complex, EMI compliant PCBs for use with radiation-hardened electronic components.
- Conducted detailed failure mode analysis, simulations, and error budget analysis to ensure robustness to conditions on the lunar surface and in orbit.
- Designed unit tester for qualification, engineering, and flight hardware validation and development.
- Implemented register-level RTOS-based embedded software for thermal control unit.

Esper Satellite Imagery

Feb 2025 – Jul 2025

Electronics Engineer

Melbourne

- Designed and validated space-grade on-board computer and power distribution system for hyperspectral Earth observation payload launching August 2026.
- Communicated with external space provider, managed interface control description documents, and conducted design reviews to ensure compliance.
- Used Atlassian project management suite to document work and organize requirements and timelines.
- Conducted extensive failure mode analysis of electrical systems.
- Designed and outsourced space-grade wiring harnesses and PCBs.

ARES Rocketry Team

Feb 2022 – Present

Chief Engineer – 2023 - 2024, Lead Systems Engineer – 2025 - Present

The University of Melbourne

- Acted as principal technical lead for a multidisciplinary team of 100 undergraduate and graduate students, to design and construct a 30,000 ft altitude sounding rocket to compete in the 2024 Spaceport America Cup. This rocket achieved 4th place out of 152 international universities.
- Wrote extensive technical documentation for project judging and to preserve knowledge for future generations of the team.
- Designed, implemented, and tested a capable flight computer for controlling rocket flight, deploying parachutes, and logging data in previous role as avionics lead. Selected for podium presentation at 2023 Spaceport America Cup (20 selected from 120 submissions).
- Currently serving as Lead systems engineer for Project Odyssey, ARES' first entirely student-designed hybrid rocket. Implementing industry-standard aerospace systems engineering practices, and acting as a chief SME for all technical aspects of rocket design.

QinetiQ Australia

Feb 2023 – Present

Technical Staff – University of Melbourne Creator Space

Melbourne

- Supervised and mentored students and staff in state-of-the-art engineering makerspace, training users from high school to professional researchers.
- Provided expert troubleshooting and education in electronics, test equipment, soldering, and PCB debugging.
- Administered technical training programs and managed competency tracking systems.
- Supported safety protocols and risk assessments for workshop operations and events.

TECHNICAL SKILLS

Circuit Design & Manufacturing

- Multi-layer PCB design and schematic capture using Altium Designer (7 years experience), KiCAD, and EagleCAD
- Extensive experience in analog, high-speed, RF, and embedded systems design.
- Design of radiation-tolerant electronic systems for LEO and lunar applications.
- Design of EMI/EMC tolerant systems to MIL-STD-461.
- SMT PCB assembly, and design for manufacture. Skilled at hand assembly of electronics.
- Circuit simulation in SPICE, TINA-TI, and QUCS.
- Use of test equipment (oscilloscopes, function generators, multimeters, spectrum analyzers, VNAs) to test and diagnose faults in circuits.

Embedded Systems & Programming

- Embedded C/C++ for microcontroller applications (5 years experience)
- Driver development for ICs and sensors
- RTOS frameworks and real-time systems
- Control system implementation (PID, Kalman filters)
- FPGA programming in Verilog (through Intel's Quartus environment)
- Communication protocols: I2C, SPI, UART, USB, CAN, Ethernet, PCIe, MIPI CSI-2
- Use of Git for version control.

Systems Engineering & Testing

- Failure mode analysis and reliability engineering
- Test campaign design and hardware validation
- Requirements management and interface control for complex systems
- Project management and leadership (have led teams of 100+ members)

AWARDS & HONORS

- 4th place out of 152 teams, Spaceport America Cup (2024)
- Podium presentation selection, Spaceport America Cup (2023)
- Dean's Honors List, The University of Melbourne (2022)
- Melbourne Global Scholar's Award (2024)
- Alan Gilbert Mobility Scholarship (2024)